

PENGWIN Task 1 v3.5 — Always-On Experts

Algorithm Description · 2026-08-05

1. Submission status

v3.5-always-expert-t075 is an upload candidate, not an automatically activated replacement. The current unified-model package remains available as the rollback model until this candidate passes the Grand Challenge container test.

Relative to v3.4, this candidate changes only Stage-B checkpoint selection: each routed anatomy always uses its Sacrum, shared-Hip, or Femur expert. Stage A, the validated hybrid target-family router, largest-component ROI policy, agglomeration threshold, and Grand Challenge input/output contract are fixed.

2. Method overview

The submission performs per-fragment instance segmentation of pelvic and femoral fractures using a serial two-stage STU-Net-B cascade:

1. **Stage A — anatomy.** A 5-class whole-volume model identifies sacrum, left/right hipbone and femur.
2. **Hybrid family routing.** A random-forest classifier selects pelvic or femur when confident. The organizers' official acquisition rule is used only when the RF is uncertain.
3. **Stage B — fracture experts.** The routed anatomy selects a Sacrum, shared-Left/Right-Hip, or Femur expert. Each emits four ABBC channels and nine learned same-instance affinity channels.
4. **Instance decoding.** Average-linkage agglomeration converts the affinity field into fragment instances and removes components below about 1 cm³.
5. **Assembly.** Instances are mapped to the official label ranges: sacrum 1-50, left hip 51-100, right hip 101-150 and femur 151-200.

3. Deployed model contract

Component	Deployed value
Stage A dataset	Dataset539_PelvicFemurAnatomyV3
Stage A trainer	PenguinTrainerSTUNetBaseAnatomyV301
Stage A fold/checkp	fold 0 / checkpoint_best.pth

Component	Deployed value
oint	
Stage A initialization	from scratch on refreshed PENGWIN anatomy data
Router	v3.3 hybrid RF, confidence margin 0.15
Stage B dataset	Dataset538_PelvicFemurBICMFragmentV5
Stage B trainers	...V308SacrumExpertDeployedVal, ...V308HipExpertDeployedVal, ..V308FemurExpertDeployedVal
Stage B fold/checkpoint	fold 0 / checkpoint_best.pth
Stage B initialization	v3.4 V308 initialized from official TotalSegmentator base_ep4k
Expert tuning	encoder frozen; decoder and heads tuned for 3 epochs
Stage B output	13 channels: 4 ABBC + 9 affinity
Decoder	decode_affinity_agglo, T=0.75

Only Stage B is TotalSegmentator-initialized. Stage A is the refreshed-data scratch model. The three packaged expert artifacts are their deterministic epoch-3 checkpoint_best.pth files.

4. Model archive layout

Grand Challenge extracts the uploaded archive directly under `/opt/ml/model`. The tarball is created with `tar -C model_payload -czf model.tar.gz .`, producing:

```
/opt/ml/model/

├─ nnunet/results/

|   └─ Dataset539_PelvicFemurAnatomyV3/
```

```
| | └─ PenguinTrainerSTUNetBaseAnatomyV301__nnUNetResEncUNetLPlans__3d_fullres/

| └─ Dataset538_PelvicFemurBICMFragmentV5/

|   └─ PenguinTrainerSTUNetBaseAffinityV308SacrumExpertDeployedVal__.../

|   └─ PenguinTrainerSTUNetBaseAffinityV308HipExpertDeployedVal__.../

|   └─ PenguinTrainerSTUNetBaseAffinityV308FemurExpertDeployedVal__.../

└─ stage1_router/stage1_target_router_fold0.joblib
```

The router artifact is serialized natively with scikit-learn 1.6.1, matching the container dependency. Its predictions were checked against the original scikit-learn 1.7.2 serialization on 4,096 deterministic inputs.

5. Local evaluation

All rows below use the same 68-patient / 132-anatomy end-to-end official-aligned proxy-v2 evaluation set.

Experiment	IoU-F	Dice	HD95 (mm)	ASD (mm)	Instance F1
v3.4 unified	0.8	0.8	6	1	0.9150
	118	507	7	8	
	28	07	5	9	
			2	1	
Selective expert OOF	0.8	0.8	6	1	0.9
	148	540	8	8	
	12	27	3	9	
			5	7	
v3.5 always	0.8	0.8	6	1	0.9
	16	55	.	.	

Experiment	IoU-F	Dice	HD95 (mm)	ASSD (mm)	Instance F1
split experts	84.2	64.9	6.87	8.52	77

Relative to unified Stage B, always-on experts improve IoU-F by 0.005013, Dice by 0.004942, HD95 by 0.065 mm, and ASSD by 0.038 mm. Precision decreases by 0.010913, instance F1 decreases by 0.002772, and split errors increase by 12. This mixed trade-off is why v3.5 is a separate hidden-score probe rather than an automatic deployment promotion. These are local proxy results, not hidden-test leaderboard results.

6. Preprocessing and inference

Images are canonicalized to LPS and processed through the nnU-Net plans used in training. Stage A predicts mutually exclusive anatomy masks. The selected anatomy ROI is converted to a CT-only bone-window crop for Stage B. Inference runs Stage A and one expert at a time on one GPU, restores the prediction to the input geometry and writes one `/output/images/peripelvic-fracture-ct-segmentation/*.mha` label map.

7. Reproducibility and limitations

- Fold 0 is used for both stages; no ensemble is deployed.
- The expert policy and $T=0.75$ were evaluated on the reported validation data, so an independent Grand Challenge container/hidden-test check is required before activation.
- Always-on experts improve overlap/surface proxies but reduce precision and instance F1 and increase fragment splitting.
- Pelvic cases switch between Sacrum and Hip expert checkpoints serially; the Grand Challenge runtime test must confirm the 10-minute case limit.
- The official challenge evaluator and hidden test distribution are not local, so the reported metrics are comparative proxies.
- Exact artifact hashes and source paths are recorded in the release `MODEL_MANIFEST.json`.

8. Runtime and licensing

The Docker image targets an NVIDIA T4 16 GiB GPU. Stage A and each Stage B expert are loaded serially to limit peak memory. STU-Net-B and nnU-Net are used under their respective open-source licenses; the Stage-B initialization comes from the official TotalSegmentator base_ep4k checkpoint.

9. References

- Isensee, F. et al. nnU-Net, *Nature Methods* 18, 203-211 (2021).
- STU-Net / TotalSegmentator bone pretraining.
- PENGWIN 2026 Task 1 baseline:
github.com/YzzLiu/PENGWIN2026_Task1_AutoSeg_Baseline.